Topological Origin of Lepton Generations: KSAU and FFGFT

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Summary

The generation problem – why exactly three particle families exist, differing only in their masses – belongs to the unsolved puzzles of the Standard Model. This study compares two independent approaches deriving masses from fundamental geometry: the *Knot-Synchronization-Adhesion Unified Theory (KSAU)* and the *Fundamental Fractal Geometric Field Theory (FFGFT)*. The synthesis reveals a deep structural correspondence: the seemingly abstract FFGFT parameters receive a direct physical interpretation through the knot topology of KSAU.

Status markers

All statements carry FFGFT status markers:
[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible **[SETZUNG]** axiomatic setting

.1 Point of Departure: The Generation Problem

The experimental mass ratios of the charged leptons:

$$m_e : m_\mu : m_\tau \approx 1 : 206.77 : 3477.4 \tag{1}$$

are described in the Standard Model by free Yukawa couplings. Neither their values nor the number of generations follow from the framework.

.2 The FFGFT Framework

The FFGFT derives all physical constants from the single dimensionless parameter

$$\xi = \frac{4}{30000} = 1.\overline{3} \cdot 10^{-4} \tag{2}$$

on a compact 4D torus T^4. The fundamental relation

$$\tilde{T} \cdot m = 1 \tag{3}$$

follows from the torus geometry. Field modes exist as standing waves with quantised winding numbers (n_θ, n_ϕ).

Resonance ladder of lepton masses

[K] The lepton masses follow the geometric ladder

$$m_i = r_i \cdot \xi^{p_i} \cdot v \tag{4}$$

with Higgs VEV $v = 246$ GeV and rational quantum numbers (r_i, p_i) (Table 1). The exponents p_i are winding numbers on a logarithmic torus with step $\Delta p = 1/3$.

Table 1: FFGFT quantum numbers of the charged leptons

Particle	Generation	n_θ	n_ϕ	$r_i = n_\phi^2/n_\theta$	p_i
Electron e	1	3	2	$4/3$	$3/2$
Muon μ	2	5	4	$16/5$	1
Tau τ	3	9	5	$25/9$	$2/3$

[K] The mass ratios are parameter-free (without v):

$$\frac{m_\mu}{m_e} = \frac{12}{5} \cdot \xi^{-1/2} \approx 207.85 \quad \text{(PDG: } 206.77,\ +0.52\,\%) \tag{5}$$

$$\frac{m_\tau}{m_\mu} = \frac{125}{144} \cdot \xi^{-1/3} \approx 16.992 \quad \text{(PDG: } 16.817,\ +1.04\,\%) \tag{6}$$

$$\frac{m_\tau}{m_e} = \frac{25}{12} \cdot \xi^{-5/6} \approx 3532 \quad \text{(PDG: } 3477,\ +1.57\,\%) \tag{7}$$

Table 2: Comparison of ladder predictions with PDG values

Ratio	Ladder (FFGFT)	PDG	Deviation
m_μ/m_e	207.85	206.77	$+0.52\,\%$
m_τ/m_μ	16.992	16.817	$+1.04\,\%$
m_τ/m_e	3532	3477	$+1.57\,\%$

[S] Falsification test: the prediction $m_\tau = 1776.97$ MeV will be decided by Belle-II (PDG: 1776.86 ± 0.12 MeV).

.3 The KSAU Theory: Topological Knots

Scientific status of the KSAU theory

[SETZUNG] The KSAU theory (Yui, Zenodo 2026) is an unpublished preprint without peer review. The knot assignment to lepton generations is a heuristic hypothesis.

- Peer-reviewed foundations: Möbius energy (Freedman et al. 1994), Avrin 2012 (*Symmetry*, MDPI), Blatt et al. 2025 (arXiv)
- Not peer-reviewed: KSAU knot assignment and mass formula

Doc. 317 uses KSAU as a **heuristic impulse**, not as established physics. All KSAU-specific statements carry **[S]**.

[S] The KSAU theory models charged leptons as stable topological knot excitations in the quantum vacuum.

Mathematical foundation: Möbius energy

[K] The Möbius energy is mathematically rigorously defined (Freedman, He & Wang 1994):

$$E_{\text{Möb}}(\gamma) = \iint \left(\frac{1}{|\gamma(s) - \gamma(t)|^2} - \frac{1}{d_\gamma(\gamma(s), \gamma(t))^2} \right) |\gamma'(s)||\gamma'(t)| \, \mathrm{d}s \, \mathrm{d}t \tag{8}$$

It is conformally invariant and possesses minimisers in every knot class. Blatt et al. (2025) prove: in every torus knot class $T(a, b)$ for co-prime a, b there exist at least two distinct critical points of the Möbius energy.

[K] Torus-knot status of the lepton assignments:

Knot	Type	Torus notation	Möbius-critical
3_1 (trefoil)	torus knot	$T(2,3)$	**yes** (Blatt et al.)
6_3 (amphicheiral)	not a torus knot	—	unknown
7_1	torus knot	$T(2,7)$	**yes** (Blatt et al.)

Important restriction: Exact analytical values $E(K)$ for 3_1, 6_3, 7_1 are not known; only numerical approximations (Kim & Kusner 1993). The KSAU mass formula requires concrete $E(K)$ values. The status of the mass prediction is therefore **[S]**.

[S] The knot structure of the leptons according to KSAU:

Table 3: KSAU knot structure of the leptons

Lepton	Knot	Knot type	Genus g	Span(Δ_K)	$E(K)/E_0$
Electron e	3_1	Trefoil	1	2	1.000
Muon μ	6_3	amphicheiral	2	4	$e^{\gamma \cdot \Delta E}$
Tau τ	7_1	torus knot	3	6	$e^{2\gamma \cdot \Delta E}$

Placement in related literature

[K] Jeon et al. (arXiv:2407.11731, 2024) demonstrate the existence of stable knot solitons in a BSM extension of the Standard Model ($\mathrm{U}(1)_{\mathrm{PQ}} \times \mathrm{U}(1)_{B-L}$, Chern-Simons coupling). Stability arises from the linking of two strings, not from Möbius energy. There is *no* direct connection to lepton masses or generations. The paper establishes that knot solitons can in principle exist in high-energy physics – nothing more.

The Faddeev-Skyrme model (Faddeev & Niemi 1997) permits knot solitons in a non-renormalisable sigma model. It is not a phenomenological model of the Standard Model.

The KSAU mass assignment is a hypothesis (Zenodo 2026) that has not appeared in any peer-reviewed publication. Doc. 317 treats it as a comparison framework with status **[SETZUNG]**, not as established physics.

The mass follows according to KSAU from the Möbius energy functional:

$$m(K) = m_0 \cdot \exp(\gamma \cdot (E(K) - E_0)), \quad \gamma = 1/e \approx 0.3679 \tag{9}$$

[K] The Alexander polynomial links the knot directly to the generation number: $\text{Span}(\Delta_K) = 2 \cdot n_g$ (Table 3).

.4 Synthesis: Structural Correspondence

Table 4: Structural correspondence KSAU ↔ FFGFT

Parameter	KSAU	FFGFT	Synthesis
Basic object	Vacuum knots $3_1, 6_3, 7_1$ **[S]**	Resonances on log. torus **[K]**	Knots on torus topology **[S]**
Generation scaling	Seifert genus $g = 1, 2, 3$	$\Delta p = 1/3$	Topological step height
Form factor	Möbius energy $E(K)$	$r_i = n_\phi^2 / n_\theta$	Geom. resonance of winding
Symmetry pivot	6_3 is amphicheiral **[S]**	$p_\mu = 1$ (integer) **[K]**	Symmetry of 2nd generation **[S]**
Chirality	Inner/outer projection **[S]**	Projector P_+ **[B]**	Origin of V–A structure **[S]**

Torus-knot bridge theorem [S]

Let $\mathcal{T}$ be a compact 4D torus with winding numbers (n_θ, n_ϕ) and $\xi = 4/30000$. Then:

1. The three lepton generations correspond to torus knots with genus $g = 1, 2, 3$.
2. The mass ratios follow the ladder $m_i = r_i \cdot \xi^{p_i} \cdot v$ with $p_i = 3/g$ and $r_i = n_\phi^2 / n_\theta$.
3. The chiral projector P_+ selects the outer state ($g_L \neq 0$, $g_R = 0$).

Status: **[S]** – the KSAU side is external (Zenodo 2026) and not verified here.

.5 Complete Quantum Number Table

Table 5: Complete quantum numbers: FFGFT ladder and KSAU topology

Lepton	Gen.	n_θ	n_ϕ	r_i	p_i	Knot	g	Span
e	1	3	2	$4/3$	$3/2$	3_1	1	2
μ	2	5	4	$16/5$	1	6_3	2	4
τ	3	9	5	$25/9$	$2/3$	7_1	3	6
ν_e	1	3	2	$4/3$	$5/2$	—	—	—
ν_μ	2	5	4	$16/5$	3	—	—	—
ν_τ	3	9	5	$25/9$	$25/9$	—	—	—

Neutrinos follow the same (n_θ, n_ϕ) structure with elevated exponents (double-ξ suppression, Doc. 006). Their KSAU knot assignments are still open.

.6 Relation to the Corpus

The lepton quantum numbers (r_i, p_i) are fully documented in Doc. 006 (particle masses, Tables 1–3) and verified by Python scripts with exact arithmetic. The FFGFT fundamental relations ($\tilde{T} \cdot m = 1$, $\xi = 4/30000$) are derived in A-series documents A080, A100, A110. The chiral projection (P_+, $g_R = 0$) is treated in A192.

[SETZUNG] The KSAU knot assignment is an external theory (Yui, Zenodo 2026), used here as a comparison framework. The structural correspondence is **[S]**.

.7 ξ as a Galois Number: Derivation from GF(3^n)

The winding numbers (n_θ, n_ϕ) of the three lepton generations (Table **??**) are fully derivable from the GF(3^n) structure **[K]**. From this, $\xi = 4/30000$ follows as a Galois number.

Winding numbers from GF(3^n)

Quantity	Value	Galois origin	
$n_{\theta,1}$	3	char(GF(3^n))	axiomatic
$n_{\phi,1}$	2	|GF$(3)^*$|	mult. group of prime field
$n_{\theta,2}$	5	$n_{\theta,1} + n_{\phi,1}$	recursion; also min. prime factor of |GF$(81)^*$| $= 80$
$n_{\phi,2}$	4	$\varphi(5)$	Euler-φ of min. prime factor
$n_{\theta,3}$	9	$n_{\theta,2} + n_{\phi,2}$	recursion; also char^2
$n_{\phi,3}$	5	min. prime factor of |GF$(81)^*$|	

Recursion law: $n_{\theta,g} = n_{\theta,g-1} + n_{\phi,g-1}$ **[K]**. The initial condition $(3, 2)$ follows from GF(3) alone. GF$(3^4)^* = 80 = 2^4 \cdot 5$ supplies the first prime factor > 3 not present at any earlier level.

Core identity and derivation of ξ

The FFGFT mass-ratio formula (Doc. 006) and the Galois group orders yield the same number:

$$\frac{(r_\mu/r_e)^2}{\xi} \;=\; |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)| \;=\; 8^2 \cdot 25 \cdot 27 \;=\; 43200 \tag{10}$$

These are not two computations of the same value but a single algebraic equation from which ξ follows:

$$\xi \;=\; \frac{(r_\mu/r_e)^2}{43200} \;=\; \frac{(12/5)^2}{43200} \;=\; \frac{1}{7500} \;=\; \frac{4}{30000} \quad [\mathrm{K}] \tag{11}$$

All factors are Galois orders: $8 = |\mathrm{GF}(9)^*|$, $27 = |\mathrm{GF}(27)|$, $5 =$ min.prime(|GF$(81)^*$|). Verification script: `pruef_331_xi_aus_galois.py`.

.8 Open Points

1. **Covariant derivatives D_μ (A192):** The chiral projection explains $SU(2)_L$ coupling qualitatively; the covariant derivatives are not yet written out explicitly.

2. **Correction** $N_0 \approx 38.6$ **(A105):** Candidate for the generation-linear correction of ladder deviations; not derived from ξ. Concerns only residual deviations.
3. **Neutrino knots:** KSAU assignments for neutrino generations not yet established.

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